Ifreet Zareef Hossain

B.Sc. in Computer Science And Engineering, AIUB



Phone: +8801872651459
Email: zareefhossain.93@gmail.com | 23-53186-3@student.aiub.edu
Address: Flat 1/C, Dominage Diamond Tower, Hazi Moron Ali Road, East Nakhalpara, Tejgaon industry area, Dhaka

SUMMARY

4th-year Computer Science and Engineering undergraduate at American International University-Bangladesh (AIUB) with hands-on experience in Object-Oriented Programming, Microcontroller and Embedded Systems, Arduino, ESP32, and Database Management. Currently developing skills in Machine Learning, with a focus on Python, fundamental ML concepts, and machine learning algorithms. Experienced in academic projects and research, including research paper publications and participation in conferences across multiple fields. Interested in Software Engineering, Embedded Systems, Artificial Intelligence, and research-oriented opportunities.

EDUCATION

B.Sc. in Computer Science And Engineering **Expected September 2027**
American International University-Bangladesh (AIUB)

- Current CGPA: 3.55

Higher Secondary Certificate (HSC) **2020**
Chowmohani Govt. S.A. College

- GPA: 4.92

Secondary School Certificate (SSC) **2018**
Noakhali Zilla School

- GPA: 4.72

CONFERENCE PAPERS & PRESENTATIONS

1. Shock-Sense: A Low-Cost Rule-Based Physiological Monitoring System for Early Hemorrhagic Shock Risk Detection
Ifreet Zareef Hossain, Mahir Mohtab Ayman, Md Sajid Hossain
IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON), IUBAT, Dhaka, 04–05 September 2026.

- Designed and developed a low-cost, rule-based physiological monitoring architecture aimed at the early detection of hemorrhagic shock risks.
- Implemented real-time data tracking logic to analyze vital signs and flag critical physiological anomalies efficiently.

2. Public Awareness and Expectations of Accelerated Reliable MRI Imaging Using Machine Learning
Ifreet Zareef Hossain, Nafisa Siddiquee, Mahir Mohtab Ayman, Prantik Baroi, Shraysree Mandal, Juena Ahmed Noshin
International Conference on Electronics and Informatics (ICEI), Bangladesh, 2026.

- Investigated public awareness, perceptions, and societal expectations regarding the integration of machine learning into accelerated reliable MRI diagnostics.
- Analyzed survey/data metrics to evaluate user trust and technological readiness for AI-driven medical imaging solutions.

3. Machine Learning in Bangladesh's Primary Education
Ifreet Zareef Hossain, Prantik Baroi, Fahim Ahmed, Dr. Fatema Tuj Zohra
International Conference on Applied Statistics and Data Science, 2025.

- Examined the feasibility, prospects, and systemic challenges of implementing machine learning and data-driven tools within Bangladesh's primary education framework.
- Proposed analytical perspectives on improving student performance tracking and resource allocation through educational data mining.

AWARDS & ACHIEVEMENTS

Hackathon Runner-up
Hack for Humanity: Hack to Protect the Environment, Powered by AI Collective (2026)

- Secured Runner-up position by developing and presenting Reroute Agro, an AI-powered agricultural marketplace addressing farmers' market-access, transportation, and post-harvest loss challenges in Bangladesh.

PROJECTS

Reroute Agro — AI-Powered Agricultural Marketplace
AI-Powered Hackathon Project (2026)

- Developed a digital agricultural marketplace connecting farmers with wholesale buyers to improve market access and selling efficiency in Bangladesh.
- Implemented a Smart Selling Advisor to compare market prices, transportation costs, spoilage risks, and expected net proceeds, helping farmers make informed selling decisions.
- Developed truck pooling and Smart Hisab features for shared transportation costs, digital invoicing, income and expense tracking, and financial record management.
- Integrated a Bangla voice assistant to make agricultural information and platform services more accessible to users.

University Management System Backend Software
Academic Software Engineering Project

- Developed a robust backend architecture to handle student enrollment, course registration, and faculty record management.
- Implemented secure database operations and API endpoints to manage data flow across multiple administrative user roles.

Airline Management System Backend Software
Object-Oriented Programming Project (C#)

- Built a comprehensive backend system in C# utilizing OOP principles for flight scheduling, seat reservations, and passenger ticketing.
- Integrated relational database connections for efficient transaction handling and data persistence.

Documentation Project: Event Planning and Vendor Management System
System Analysis and Design

- Prepared detailed system requirement specifications (SRS), use-case diagrams, and process workflows for an automated event planning and vendor coordination platform.
- Structured documentation covering risk assessment, system architecture, and stakeholder management frameworks.

TRAINING AND CERTIFICATION

- **Fundamental of ML** – Gobeshona Learning (2025)
- **Object-Oriented Programming with Java** – Udemy (2025)
- **Front-End Development** – Udemy (2025)
- **Digital Marketing** – SR Dream IT (2024)
- **Master in JavaScript** – Coursera (2025)

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, JavaScript, C, C#
- **Web Development:** HTML, CSS, Front-End Development, REST APIs
- **Machine Learning & Data:** Fundamental Machine Learning, Data Analysis
- **Embedded Systems & Hardware:** Arduino, ESP32, Microcontrollers, NI Multisim, Sensor Integration
- **Software Engineering & Concepts:** Object-Oriented Programming (OOP), Software Development Life Cycle (SDLC), Git, GitHub
- **Simulation & Tools:** MATLAB, VS Code, Database Management (MySQL)

VOLUNTEER EXPERIENCE

Event Volunteer **2026**
Seminar on Semiconductor Device Modeling and COMSOL Multiphysics

- Handled crowd management, coordinated workshop sessions, and guided attendees during the technical seminar.

Marathon Kit Volunteer & Race Track Handler **2025**
Activepulse Bangladesh Marathon

- Managed marathon kit distribution and assisted in race track coordination to ensure smooth event operations.

Campus Ambassador **2021–2022**
National University Skill Development Forum (NUSDF)

- Acted as a campus representative and inspired students to engage in academic research and skill development activities.

Campus Ambassador **2020–2021**
International Model United Nations (IMUN), Indonesia

- Represented the institution across various networks and promoted educational courses and programs.

REFERENCES

1. Md Sajid Hossain
BSc (EEE), MIEB, MIEEE
Senior Assistant Professor
Department of Electric and Electronics Engineering, Faculty of Engineering
American International University-Bangladesh
Email: sajidh@aiub.edu
Phone: 01670381831

2. Tanvir Hossain
Assistant Vice Principal & Senior Manager, Corporate South
One Bank PLC (Head Office)
Email: Tanvirnomi.onebank@gmail.com
Phone: 01717552215

3. Zobaer Hossain
Assistant Manager (Engineer), Tasnin Chemicals
Concern of Meghna Group of Industry
Email: zobaerhossain@gmail.com
Phone: +8801670-744-353